UNIT-2

Linear differencial equations of higher Order.

Assignment - 2

1. $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$, given that $y(0) = 0$, $y'(0) = 15$

Sol: Given D.E is $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$; $y(0) = 0$

$y'(0) = 15$

Operator form

$$(D^2 + 5D + 6)y = 0$$

auxillary eqn is $m^2 + 5m + 6 = 0$

$$m^2 + 2m + 3m + 6 = 0$$

$$m(m+2) + 3(m+2) = 0$$

$$m = -2, -3$$

$$CF = c_1 e^{-2x} + c_2 e^{-3x}$$

$PI = 0$ because $Q(x) = 0$

General Solution $y = CF + PI$

$$y = c_1 e^{-3x} + c_2 e^{-2x} \rightarrow (1)$$

$$y(0) = 0$$

$$x = 0, y = 0$$

$$0 = c_1 e^{-3(0)} + c_2 e^{-2(0)}$$

$$0 = c_1 + c_2$$

$$\boxed{c_1 = -c_2}$$

$$y' = C_1 e^{-3x}(-3) + C_2 e^{-2x}(-2)$$

$$y' = -3C_1 e^{-3x} - 2C_2 e^{-2x}$$

$$y'(0) = 15$$

$$x = 0, \quad y = 15$$

$$15 = -3C_1 e^{-3(0)} - 2C_2 e^{-2(0)}$$

$$15 = -3C_1 - 2C_2 \qquad [-C_2 = C_1]$$

$$15 = -3C_1 + 2(C_1)$$

$$15 = -C_1$$

$$\boxed{C_1 = -15} \qquad C_2 = -C_1$$

$$= -(-15)$$

$$\boxed{C_2 = 15}$$

Now, G.S is $y = -15e^{-3x} + 15e^{-2x}$

(2) Solve $\frac{d^4x}{dt^4} + 4x = 0$

Sol:- Given D.E is $\frac{d^4x}{dt^4} + 4x = 0$

→ operator form $f(D).x = 0$

$$(D^4 + 4).x = 0$$

→ auxillary eqn $m^4 + 4 = 0$

add and sub $4m^2$ in auxillary eqn

$$m^4 + 4m^2 + 4 - 4m^2 = 0$$

$$(m^2+2)^2 - (2m)^2 = 0$$

$$a^2 - b^2 = (a-b)(a+b)$$

$$(m^2 + 2 - 2m)(m^2 + 2 + 2m) = 0$$

$$(m^2 - 2m + 2)(m^2 + 2m + 2) = 0$$

$m^2 - 2m + 2 = 0$

$$m = \frac{2 \pm \sqrt{4-8}}{2}$$

$$= \frac{2 \pm 2i}{2}$$

$$m = 1 \pm i$$

$m^2 + 2m + 2 = 0$

$$m = \frac{-2 \pm \sqrt{4-8}}{2}$$

$$= \frac{-2 \pm 2i}{2}$$

$$= -1 \pm i$$

$$CF = e^t (c_1 \cos t + c_2 \sin t) + e^{-t} (c_3 \cos t + c_4 \sin t)$$

G.S is

$x = CF + PI$ $\quad PI = 0$ because $Q(x) = 0$.

$$x = e^t (c_1 \cos t + c_2 \sin t) + e^{-t} (c_3 \cos t + c_4 \sin t)$$

③ $(D^2 - 6D + 9)\, y = 6e^{3x} + 7e^{-2x} - \log 2$

Sol: Auxillary eqn $f(m) = 0$

$$m^2 - 6m + 9 = 0$$

$$m^2 - 3m - 3m + 9 = 0$$

$$m(m-3) - 3(m-3) = 0$$

$$m = 3, 3$$

$$CF = (C_1 + C_2 x)\, e^{3x}$$

$$PI = \frac{1}{f(D)} \cdot Q(x)$$

$$= \frac{1}{D^2 - 6D + 9} \left(6e^{3x} + 7e^{-2x} - \log 2 \right)$$

$$PI = \left[\frac{1}{D^2 - 6D + 9} \cdot 6e^{3x} \right] + \left[\frac{1}{D^2 - 6D + 9} \cdot 7e^{-2x} \right] - \left[\frac{1}{D^2 - 6D + 9} \log 2 \right]$$

$$PI = PI_1 + PI_2 - PI_3 \longrightarrow (1)$$

Now, $PI_1 = 6 \left[\frac{1}{D^2 - 6D + 9} e^{3x} \right]$

$a = 3$, $f(D) = D^2 - 6D + 9$

$$f(D = a) = (3)^2 - 6(3) + 9$$

$$= 9 - 18 + 9$$

$$= 0$$

Now, $PI_1 = 6\left[x \cdot \frac{1}{\frac{d}{dD} f(D)} \cdot e^{3x}\right]$

$= x \cdot \left[\frac{1}{2D-6} \cdot e^{3x}\right]$

$a = 3, \quad f(D) = 2D - 6 \quad \Rightarrow \frac{1}{f(D=a)} \cdot e^{3x}$

$f(D=a) = 2(3) - 6$

$= 6 - 6 = 0$

$PI_1 = 6x \cdot x\left[\frac{1}{\frac{d}{dD} f(D)} \cdot e^{3x}\right]$;

$= \overset{3}{\cancel{6}} x^2 \frac{1}{\cancel{2}} \cdot e^{3x}$

$\boxed{PI_1 = 3x^2 e^{3x}}$

Now, $PI_2 = 7 \cdot \left[\frac{1}{D^2 - 6D + 9} \cdot e^{-2x}\right]$

$a = -2, \quad f(D) = D^2 - 6D + 9$

$f(D=a) = (-2)^2 - 6(-2) + 9$

$= 4 + 12 + 9$

$= 25 \quad \Rightarrow 7 \cdot \frac{1}{f(D=a)} \cdot e^{-2x}$

$PI_2 = 7 \cdot \frac{1}{25} e^{-2x}$

$\boxed{PI_2 = \frac{7}{25} e^{-2x}}$

$$PI_3 = \frac{1}{D^2-6D+9} \cdot \log 2$$

$$= \log 2 \cdot \left[\frac{1}{D^2-6D+9} \cdot e^{0x} \right]$$

$a = 0$, $f(D) = D^2-6D+9$

$$f(D=a) = 0-0+9$$

$$= 9$$

$$PI_3 = \log 2 \cdot \frac{1}{f(D=a)} \cdot e^{0x}$$

$$= \log 2 \cdot \frac{1}{9} \cdot (1)$$

$$\boxed{PI_3 = \frac{\log 2}{9}}$$

$$PI = 3x^2 e^{3x} + \frac{7}{25} e^{-2x} - \frac{\log 2}{9}.$$

G.S is $y = CF + PI$

$$y = (C_1 + C_2 x) e^{3x} + 3x^2 e^{3x} + \frac{7}{25} e^{-2x} - \frac{\log 2}{9}.$$

(4) Solve $(D^2+5D-6)\, y = \sin 4x \sin x$.

Sol: auxillary eqn $m^2+5m-6 = 0$

$m^2 + 6m - m - 6 = 0$

$m(m+6) - 1(m+6) = 0$

$(m-1)(m+6) = 0$

$m = 1, \quad m = -6$

$CF = C_1 e^{x} + C_2 e^{-6x}$.

PI $\quad Q(x) = \sin 4x \sin x$.

w.k.t,

$2 \sin A \sin B = \cos(A-B) - \cos(A+B)$.

$\sin A \sin B = \frac{1}{2}\left[\cos(A-B) - \cos(A+B)\right]$

$\sin 4x \sin x = \frac{1}{2}\left[\cos 3x - \cos 5x\right]$

Now,

$$PI = \frac{1}{f(D^2 = -b^2)} \cdot \frac{1}{2}\left[\cos 3x - \cos 5x\right].$$

$$PI = \frac{1}{2}\left[\frac{1}{f(D^2=-b^2)} \cdot \cos 3x - \frac{1}{f(D^2=-b^2)} \cos 5x\right].$$

$\frac{1}{f(D^2=-b^2)} \cdot \cos 3x \rightarrow PI_1$ $\qquad$ $\frac{1}{f(D^2=-b^2)} \cos 5x \rightarrow PI_2$

$$PI_1 = \frac{1}{f(D^2=-b^2)} \cos 3x$$

$-b^2 = -(3)^2 = -9$ $\qquad$ $f(D) = D^2 + 5D - 6$

$= -9 + 5D - 6$

$= 5D - 15 = 5(D-3)$

$$PI_1 = \frac{1}{5} \cdot \frac{1}{D-3} \cos 3x$$

$$= \frac{D+3}{5} \cdot \frac{\cos 3x}{D^2 - 3^2}$$

$$= \frac{1}{5} \frac{1}{D^2-(3)^2} (D(\cos 3x) + 3\cos 3x)$$

$$D^2 = -b^2 \Rightarrow D^2 = -9$$

$$= \frac{1}{5} \frac{1}{-9-9} (-\sin 3x\,(3) + 3\cos 3x)$$

$$= \frac{3}{(5)(-18)_{6}} (\cos 3x - \sin 3x)$$

$$= \frac{-1}{30} \cos 3x - \sin 3x \Rightarrow \frac{1}{30} \sin 3x - \cos 3x$$

$$\boxed{PI_1 = \frac{1}{30} \sin 3x - \cos 3x}$$

$$PI_2 = \frac{1}{f(D^2 = -b^2)} \cos 5x$$

$b = 5$ $\quad -b^2 = -25$

$$f(D) = D^2 + 5D - 6$$
$$= -25 + 5D - 6$$
$$= 5D - 31$$

$$PI_2 = \frac{1}{5D-31} \cos 5x$$

$$= \frac{5D+31}{(5D)^2 - (31)^2} \cdot \cos 5x.$$

$$= \frac{5\,D(\cos 5x) + 31\cos 5x}{25(D^2) - 961}$$

$D^2 = -b^2 \Rightarrow -b^2 = -25$

$$= \frac{5\,(-\sin 5x(5)) + 31\cos 5x}{25(-25) - 961}$$

$$= \frac{-25\sin 5x + 31\cos 5x}{-625 - 961}$$

$$= \frac{31\cos 5x - 25\sin 5x}{-1586}$$

$$\boxed{PI_2 = \frac{25\sin 5x - 31\cos 5x}{1586}}$$

$$PI = \frac{1}{2}\left[\frac{1}{30}\sin 3x - \cos 3x\right] - \frac{1}{2}\left[\frac{25\sin 5x - 31\cos 5x}{1586}\right]$$

$$PI = \frac{1}{60}(\sin 3x - \cos 3x) - \frac{1}{3172}(25\sin 5x - 31\cos 5x).$$

G.S is $y = CF + PI$

$$y = c_1 e^{x} + c_2 e^{-6x} + \frac{1}{60}(\sin 3x - \cos 3x) - \frac{1}{3172}(25\sin 5x - 31\cos 5x)$$

⑤ Solve $(D^3 + 4D)y = 5 + \sin 2x + 3^x$

Sol: Auxillary eqn $m^3 + 4m = 0$

$m(m^2 + 4) = 0$

$m = 0 \qquad m^2 = -4$

$m = \pm 2i$

$m = 0, \pm 2i$

$CF = C_1 e^{0x} + C_2 \cos 2x + C_3 \sin 2x$

$PI = \frac{1}{f(D)} \cdot (5 + \sin 2x + 3^x)$

$PI = \frac{5}{f(D)} + \frac{\sin 2x}{f(D)} + \frac{3^x}{f(D)}$

$\downarrow PI_1 \qquad \downarrow PI_2 \qquad \downarrow PI_3$

Now,

$PI_1 = 5\left[\frac{1}{f(D)} e^{0x}\right]$

Type - I

$a = 0 \qquad f(D) = D^3 + 4D$

$f(D=a) = 0 + 0$

$= 0$

$PI_1 = 5 \cdot x \frac{1}{\frac{d}{dD} f(D)} e^{0x}$

$= 5x \frac{1}{3D^2 + 4} \cdot e^{0x}$

$a = 0, \quad f(0) = 3(0) + 4$

$= 4$

$PI_1 = 5x \cdot \frac{1}{4} \cdot e^{0x}$

$$\boxed{PI_1 = \frac{5}{4}x}$$

Now, $PI_2 = \frac{1}{f(D)} \cdot \sin 2x$

Type - II

$b = 2$

$D^2 = -b^2$

$D^2 = -4$

$f(D) = D^3 + 4D$

$= D^2 \cdot D + 4D$

$= -4D + 4D$

$= 0$

$$PI_2 = x \cdot \frac{1}{\frac{d}{dD} f(D)} \cdot \sin 2x$$

$$= x \cdot \frac{1}{3D^2 + 4} \cdot \sin 2x$$

$$= x \cdot \frac{1}{3(-4) + 4} \cdot \sin 2x$$

$$= x \cdot \frac{1}{-8} \sin 2x$$

$$\boxed{PI_2 = -\frac{x}{8} \sin 2x}$$

$$PI_3 = \frac{1}{f(D)} \cdot 3^x = \frac{1}{f(D)} \cdot e^{(\log 3)x}$$

Type - I

$a = \log 3$

$f(D) = D^3 + 4D$

$f(D=a) = (\log 3)^3 + 4(\log 3)$

$\log 3 = \ln(3)$
$= 1.09$

$= (1.09)^3 + 4(1.09)$

$= 1.29 + 4.36$

$= 5.65$

$PI_3 = \frac{1}{5.65} e^{\log 3x}$

$\boxed{PI_3 = \frac{1}{5.65} e^{1.09x}}$

$PI = \frac{5}{4}x - \frac{x}{8}\sin 2x + \frac{1}{5.65} e^{1.09x}$

G.S is $y = CF + PI$

$y = c_1 e^{0x} + c_2 \cos 2x + c_3 \sin 2x + \frac{5}{4}x - \frac{x}{8}\sin 2x + \frac{e^{1.09x}}{5.65}$

⑥ Solve $\frac{d^2y}{dx^2} + \frac{dy}{dx} = x^2 + 2x + 4$

Sol: Given $\frac{d^2y}{dx^2} + \frac{dy}{dx} = x^2 + 2x + 4$

Operator form $(D^2 + D)y = x^2 + 2x + 4$

Auxillary eqn $m^2 + m = 0$

$m(m+1) = 0$

$m = 0, -1$

CF is $CF = c_1 e^{0x} + c_2 e^{-x}$

$\boxed{CF = c_1 + c_2 e^{-x}}$

$$PI = \frac{1}{f(D)} \cdot (x^2+2x+4)$$

$$f(D) = D^2+D$$

$$= D[1+D]$$

$$PI = \frac{1}{D(1+D)} (x^2+2x+4)$$

$$= \frac{(1+D)^{-1}}{D} (x^2+2x+4)$$

W.K.T

$$(1+t)^{-1} = 1-t+t^2-t^3+\cdots$$

$$PI = \frac{1}{D}[1-D+D^2-D^3+\cdots](x^2+2x+4)$$

$$= \frac{1}{D}[(x^2+2x+4)-D(x^2+2x+4)+D^2(x^2+2x+4)-\cdots]$$

$$= \frac{1}{D}[x^2+2x+4-2x-2+2]$$

$$= \int (x^2+4)\,dx = \frac{x^3}{3}+4x$$

G.S is $y = CF + PI$

$$y = C_1 + C_2 e^{-x} + \frac{x^3}{3} + 4x$$

$$(D^2-4D+4)y = e^{2x} + 8x^2$$

Auxillary eqn $m^2-4m+4=0$

$$m^2 - 2m - 2m + 4 = 0$$

$$m(m-2) - 2(m-2) = 0$$

$$m = 2, 2$$

$$CF = (C_1 + C_2 x)\, e^{2x}$$

$$PI = \frac{1}{f(D)} \cdot (e^{2x} + 8x^2)$$

$$= \left[\frac{1}{f(D)} \cdot e^{2x}\right] + 8 \cdot \left[\frac{1}{f(D)} \cdot x^2\right]$$

PI_1 $\qquad$ PI_2

Now, $PI_1 = \frac{1}{f(D)} \cdot e^{2x}$

$a = 2, \quad f(D) = D^2 - 4D + 4$

$f(D=a) = 4 - 4(2) + 4$

$= 0$

$$PI_1 = x\left[\frac{1}{\frac{d}{dD} f(D)} \cdot e^{2x}\right]$$

$$= x \cdot \frac{1}{2D - 4} \cdot e^{2x}$$

$$= x \cdot \frac{1}{2(2) - 4} \cdot e^{2x}$$

$$PI_1 = x\,(0) \cdot e^{2x}$$

$$PI_1 = x \cdot x \cdot \frac{1}{\frac{d}{dD} f(D)} \cdot e^{2x}$$

$$= x^2 \frac{1}{2} e^{2x}$$

$$\boxed{PI_1 = \frac{x^2 e^{2x}}{2}}$$

$$PI_2 = \frac{1}{f(D)} \cdot x^2$$

$$f(D) = D^2 - 4D + 4$$

$$= 4\left[1 + \left(\frac{D^2 - 4D}{4}\right)\right]$$

$$PI_2 = \frac{1}{4 \cdot \left(1 + \frac{D^2 - 4D}{4}\right)} \cdot x^2$$

$$= \frac{1}{4} \cdot \left[1 + \left(\frac{D^2 - 4D}{4}\right)\right]^{-1} \cdot x^2$$

$$= \frac{1}{4}\left[1 - \left(D - \frac{D^2}{4}\right)\right]^{-1} \cdot x^2$$

WKT, $(1-t)^{-1} = 1 + t + t^2 + t^3 + \cdots$

$$PI_2 = \frac{1}{4}\left[1 + \left(D - \frac{D^2}{4}\right) + \left(D - \frac{D^2}{4}\right)^2 \cdots\right] \cdot x^2$$

$$PI_2 = \frac{1}{4}\left[1 + D - \frac{D^2}{4} + D^2 \cdots\right] \cdot x^2$$

$$PI_2 = \frac{1}{4}\left[1 + D + \frac{3}{4}D^2 \cdots\right] \cdot x^2$$

$$PI_2 = \frac{1}{4}\left[x^2 + 2x + \frac{3}{4}(2) \cdots\right]$$

$$\boxed{PI_2 = \frac{1}{4}\left[x^2 + 2x + \frac{3}{2}\right]}$$

$$PI = PI_1 + PI_2$$

$$= \frac{x^2 e^{2x}}{2} + \frac{8}{4}\left[x^2 + 2x + \frac{3}{2}\right]$$

$$= \frac{x^2 e^{2x}}{2} + 2\left(x^2 + 2x + \frac{3}{2}\right)$$

(8) $(D^2 - 1)y = x \sin 3x + \cos x$

Sol:- Auxillary eqn $m^2 - 1 = 0$

$$m^2 = 1$$

$$\boxed{m = \pm 1}$$

$$CF = C_1 e^{-x} + C_2 e^{x}$$

$$PI = \frac{1}{f(D)} \cdot (x \sin 3x + \cos x)$$

$$PI = \frac{1}{D^2-1}(x \sin 3x) + \frac{1}{D^2-1}\cos x$$

$$\downarrow \qquad\qquad \downarrow$$

$$PI_1 \qquad\qquad PI_2$$

$$PI_1 = \frac{1}{D^2-1}(x).\sin 3x$$

$$= \frac{1}{f(D)}(xv) = \left[x - \frac{f'(D)}{f(D)}\right]\cdot\frac{v}{f(D)}$$

$$PI_1 = \left[x - \frac{2D}{D^2-1}\right]\frac{\sin 3x}{D^2-1}$$

Now, $D^2 = -b^2 \Rightarrow -9$

$$PI_1 = \left[x - \frac{2D}{-9-1}\right]\left[\frac{\sin 3x}{-9-1}\right]$$

$$= \left(x + \frac{D}{5}\right)\left(\frac{\sin 3x}{-10}\right)$$

$$PI_1 = \left(x + \frac{D}{5}\right)\left[-\frac{1}{10}\sin 3x\right] = -\frac{x}{10}\sin 3x - \frac{3}{50}\cos 3x$$

$$PI_2 = \frac{1}{D^2-1}\cos x$$

$D^2 = -b^2 \Rightarrow -1$

$$PI_2 = \frac{1}{-1-1}\cos x$$

$$= -\frac{1}{2}\cos x$$

G.S is $y = CF + PI$

$$y = c_1 e^{-x} + c_2 e^{x} - \frac{x}{10}\sin 3x - \frac{3}{50}\cos 3x - \frac{1}{2}\cos x$$

⑨ Solve $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = x^2 \cdot e^{-3x}$

Sol:- Given $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = x^2 \cdot e^{-3x}$

operator form $(D^2 + 2D - 3) \cdot y = x^2 \cdot e^{-3x}$

auxillary eqn $m^2 + 2m - 3 = 0$

$m^2 + 3m - m - 3 = 0$

$m(m+3) - 1(m+3) = 0$

$(m+3)(m-1) = 0$

$m = -3, 1$

$CF = c_1 e^{x} + c_2 e^{-3x}$

$PI = \frac{1}{f(D)} (e^{ax} \cdot v) = e^{ax} \cdot \frac{1}{f(D+a)} \cdot v$

$\boxed{a = -3}$

$f(D+a) = f(D-3) = (D-3)^2 + 2(D-3) - 3$

$= D^2 + 9 - 6D + 2D - 6 - 3$

$= D^2 - 4D$

$PI = e^{-3x} \cdot \left[\frac{1}{D^2 - 4D} x^2 \right]$.

Now, $\frac{1}{D^2 - 4D} = \frac{1}{4D(D/4D - 1)} = \frac{1}{4D\left[\frac{D}{4} - 1\right]}$

$$PI = \frac{-e^{-3x}}{4D}\left[1-\frac{D}{4}\right]^{-1}(x^2)$$

WKT: $(1-t)^{-1} = 1+t+t^2+t^3+\cdots\cdots$

$$PI = \frac{-e^{-3x}}{4D}\left[1+\frac{D}{4}+\frac{D^2}{16}\right](x^2)$$

$$= \frac{-e^{-3x}}{4D}\left[x^2+\frac{1}{4}(2x)+\frac{1}{16}(2)\right]$$

$$= \frac{-e^{-3x}}{4D}\left[x^2+\frac{x}{2}+\frac{1}{8}\right]$$

$$= \frac{-e^{-3x}}{4}\left[\int x^2\,dx+\frac{1}{2}\int x\,dx+\frac{1}{8}\int dx\right]$$

$$= \frac{-e^{-3x}}{4}\left[\frac{x^3}{3}+\frac{1}{2}\cdot\frac{x^2}{2}+\frac{1}{8}x\right]$$

$$= \frac{-e^{-3x}}{4}\left[\frac{x^3}{3}+\frac{x^2}{4}+\frac{x}{8}\right]$$

G.S is $y = CF + PI$

$$y = c_1e^{x}+c_2e^{-3x}-\frac{e^{-3x}}{4}\left[\frac{x^3}{3}+\frac{x^2}{4}+\frac{x}{8}\right]$$

(10) $\frac{d^2y}{dx^2}-2\frac{dy}{dx}+y = x\cdot e^{x}\cdot\sin x$

Sol:- Given D.E is $\frac{d^2y}{dx^2}-2\frac{dy}{dx}+y = x\cdot e^{x}\cdot\sin x$

Operator form $(D^2-2D+1)y=0$

Auxillary form $m^2-2m+1=0$

$$(m-1)^2=0$$

$$m=1,1$$

$$CF=(C_1+C_2x)e^x$$

$$P.I=\frac{1}{f(D)}\cdot x\cdot e^x\cdot\sin x=\frac{1}{(D-1)^2}(xe^x\sin x)$$

$$PI=\frac{1}{f(D)}e^{ax}V(x)=e^{ax}\left[\frac{1}{f(D+a)}V(x)\right]$$

$$=e^x\left[\frac{1}{((D+1)-1)^2}\sin x\right]$$

$$=e^x\frac{1}{D^2}x\sin x \neq e^x\iint x\sin x\,dx\,dx.$$

$$=e^x\int x(\cos x)\,dx=-e^x\sin x.$$

$$=e^x\iint x\cdot\sin x\,dx\,dx.$$

$$=e^x\int(-x\cos x+\sin x)\,dx$$

$$=e^x[-x\sin x-2\cos x].$$

$$PI=-e^x[x\sin x+2\cos x]$$

G.S is $y = CF + PI$

$$y = (c_1 + c_2 x)e^x - e^x [x \sin x + 2\cos x]$$

11. $(D^2 - 4D + 4)y = 8x^2 e^{2x} \sin 2x$

Sol: Given D.E is $(D^2 - 4D + 4)y = 8x^2 e^{2x} \sin 2x$

Auxillary eqn is $m^2 - 4m + 4 = 0$

$$m^2 - 2m - 2m + 4 = 0$$

$$m(m-2) - 2(m-2) = 0$$

$$(m-2)(m-2) = 0$$

$$m = 2, 2$$

$$CF = (c_1 + c_2 x)e^{2x}$$

$$PI = \frac{1}{f(D)} (16^2 + 4DA/4)$$

$$PI = \frac{1}{f(D)} \cdot 8 \cdot x^2 \cdot e^{2x} \cdot \sin 2x = \frac{1}{(D-2)^2} [8x^2 e^{2x} \sin(2x)]$$

Cosine

$$PI = 8e^{2x} \frac{1}{((D+2)-2)^2} [x^2 \cdot \sin(2x)]$$

$$PI = 8e^{2x} \frac{1}{D^2} [x^2 \sin 2x]$$

$$PI = 8e^{2x} \iint (x^2 \cdot \sin 2x)\, dx\, dx$$

First Integration

$$\int x^2 \sin(2x)\,dx = -\frac{x^2}{2}\cos(2x) + \frac{x}{2}\sin(2x) + \frac{1}{4}\cos(2x)$$

Second Integration

$$\int -\frac{x^2}{2}\cos(2x)\,dx + \int \frac{x}{2}\sin(2x) + \int \frac{1}{4}\cos 2x \Rightarrow$$

$$\Rightarrow -\frac{1}{4}x^2\sin(2x) - \frac{1}{2}x\cos(2x) + \frac{3}{8}\sin(2x)$$

$$P.I = 8e^{2x}\left[-\frac{1}{4}x^2\sin 2x - \frac{1}{2}x\cos 2x + \frac{3}{8}\sin 2x\right]$$

$$P.I = e^{2x}\left[-2x^2\sin(2x) - 4x\cos(2x) + 3\sin(2x)\right]$$

G.S is $y = CF + PI$

$$y = (c_1 + c_2 x)e^{2x} + e^{2x}\left[-2x^2\sin(2x) - 4x\cos(2x) + 3\sin 2x\right]$$

(12) Solve $y'' - 6y' + 9y = \frac{e^{3x}}{x^2}$ by the method of variation of parameters.

Sol:- Given, $y'' - 6y' + 9y = \frac{e^{3x}}{x^2}$

$$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = \frac{e^{3x}}{x^2}$$

operator form $(D^2 - 6D + 9)y = \frac{e^{3x}}{x^2}$

Auxillary eqn $m^2 - 6m + 9 = 0$

$m^2 - 3m - 3m + 9 = 0$

$m(m-3) - 3(m-3) = 0$

$m = 3, 3$

$CF = (C_1 + C_2 x)\, e^{3x} = C_1 e^{3x} + C_2 x e^{3x}$

$u = e^{3x}$; $v = x \cdot e^{3x}$

$$w = \begin{vmatrix} u & v \\ u' & v' \end{vmatrix} = \begin{vmatrix} e^{3x} & x \cdot e^{3x} \\ 3e^{3x} & x e^{3x}(3) + e^{3x}(1) \end{vmatrix}$$

$= e^{6x} + 3x e^{6x} - 3x e^{6x}$

$= e^{6x}$

$$A = -\int \frac{Q \cdot v}{w} \cdot dx$$

$$= -\int \frac{e^{3x}}{x^2} \cdot \frac{x \cdot e^{3x}}{e^{3x} \cdot e^{3x}} \cdot dx$$

$$= -\int \frac{1}{x}\, dx = -\log x$$

$A = -\log x$

$$B = \int \frac{Q \cdot u}{w}\, dx$$

$$= \int \frac{e^{3x}}{x^2} \cdot \frac{e^{3x}}{e^{3x} \cdot e^{3x}} \cdot dx = \int \frac{1}{x^2}\, dx$$

$= -\frac{1}{x}$

$PI = Au + Bv$

$= \log x\,(e^{3x}) + \frac{-1}{x}\,(x \cdot e^{3x})$

$= -\log x\, e^{3x} - e^{3x}$

$= -e^{3x}\,(\log x + 1)$

$\therefore$ G.S is $y = CF + PI$

$y = c_1 e^{3x} + c_2 x e^{3x} - e^{3x}(\log x + 1)$

(13) Solve $\frac{d^2y}{dx^2} + a^2 y = \tan ax$ by the method of variation of parameters.

Sol: Given D.E $\frac{d^2y}{dx^2} + a^2 y = \tan ax$

Operation form $(D^2 + a^2)\,y = \tan ax$

Auxillary form $(m^2 + a^2)\,y = \tan ax$

$m = \pm ai$

$CF = e^{0x}\,[c_1 \cos ax + c_2 \sin ax]$

$CF = c_1 \cos ax + c_2 \sin ax$

$u = \cos ax$

$v = \sin ax$

$$W = \begin{vmatrix} u & v \\ u' & v' \end{vmatrix} = \begin{vmatrix} \cos ax & \sin ax \\ -a\sin ax & a\cos ax \end{vmatrix}$$

$$= a\cos^2 ax + a\sin^2 ax = a(1) = a$$

$$\boxed{W = a}$$

$$PI = Au + Bv$$

$$A = -\int \frac{Q(x).v}{W}\,dx = -\int \frac{\tan ax \cdot \sin ax}{a}\,dx$$

$$= \frac{-1}{a}\int \frac{\sin ax}{\cos ax} \cdot \sin ax \cdot dx$$

$$= \frac{-1}{a}\int \frac{\sin^2 ax}{\cos ax} \cdot dx = \frac{-1}{a}\int \frac{1-\cos^2 ax}{\cos ax}\,dx$$

$$= \frac{-1}{a}\left[\int (\sec ax - \cos ax)\,dx\right]$$

$$= \frac{-1}{a}\left[\frac{1}{a}(\log|\sec ax + \tan ax|) - \frac{\sin ax}{a}\right]$$

$$= \frac{-1}{a^2}\left[\log|\sec ax + \tan ax| - \sin ax\right]$$

$$B = \int \frac{Q(x).u}{W}\,dx = \int \frac{\tan ax \cdot \cos ax}{a}\,dx$$

$$= \frac{1}{a}\int \sin ax \cdot dx$$

$$= \frac{1}{a}\left[\frac{-\cos ax}{a}\right] = \frac{-\cos ax}{a^2}$$

$$PI = Au + Bv$$

$PI = AU + BV$

$= \left(\frac{-1}{a^2}\log|\sec ax + \tan ax| + \frac{1}{a}\sin ax\right)\cos ax +$

$\left(-\frac{\cos ax}{a^2}\right)\sin ax$

$= \frac{-1}{a^2}\log|\sec ax + \tan ax| + \frac{1}{a^2}\sin ax \cos ax$

$-\frac{1}{a^2}\sin ax \cos ax$

$= -\frac{1}{a^2}\log|\sec ax + \tan ax|$

G.S is $y = CF + PI$

$y = c_1 \cos ax + c_2 \sin ax - \frac{1}{a^2}\log|\sec ax + \tan ax|.$

out off

(14) Solve $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = \log x \sin(\log x)$

Sol: This is a Cauchy-Euler homogeneous linear differencial equation. To solve it, we convert it into a linear differencial equation with constant coefficients using a substitution.

Given, $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = \log x \cdot \sin(\log x)$

let $x = e^z$, which implies $z = \log x$

The derivatives transform as follows:

1. $x \frac{dy}{dx} = Dy$ 2. $x^2 \frac{d^2y}{dx^2} = D(D-1)y$

where $D = \frac{d}{dz}$

Substituting these into the Original equation:

$$D(D-1)y + Dy + y = z \sin z$$

$$(D^2 - D + D + 1)y = z \sin z$$

$$(D^2 + 1)y = z \sin z$$

auxillary form $m^2 + 1 = 0$

$$m^2 = -1$$

$$\boxed{m = \pm i}$$

$$CF = C_1 \cos z + C_2 \sin z$$

Substituting $z = \log x$;

Now, $CF = C_1 \cos(\log x) + C_2 \sin(\log x)$

$$PI = \frac{1}{D^2+1}(z \sin z)$$

using $x.v$ rule of PI,

Here V is a function z ;

$$PI = \frac{1}{f(D)} \cdot e^{ax}$$

$$PI = e$$

$$PI = \frac{1}{f(D)} (xv) \Longrightarrow \left[x - \frac{f'(D)}{f(D)} \right] \frac{v}{f(D)}$$

Now, $$PI = \left[z - \frac{1}{D^2+1} (2D) \right] \frac{\sin z}{D^2+1}$$

Here $\frac{\sin z}{D^2+1}$ II nd Type

$b = 1 \Rightarrow D^2 = -b^2 = -1 \rightarrow \frac{\sin z}{-1+1} = 0$.

Then

$$PI = x \cdot \frac{1}{\frac{d}{dD} f(D)} \sin bx \cdot \quad [\text{Here } x = z]$$

Now,

$$.PI = \left[z - \frac{2D}{D^2+1} \right] \left[-\frac{z \cos z}{2} \right]$$

$$PI = -\frac{z^2 \cos z}{2} + \frac{D}{D^2+1} (z \cos z)$$

$$PI = -\frac{z^2 \cos z}{2} + \frac{\cos z - z \sin z}{D^2+1}$$

we know $(D^2+1)\sin z = 0$

So the Operator $\frac{1}{D^2+1}$ on $\cos z$ and $\sin z$ gives terms multiplied by z

Thus

$$\frac{1}{D^2+1}\cos z = \frac{z}{2}\sin z \qquad [\text{Type - II}]$$

$$\frac{1}{D^2+1} z\sin z = \frac{z^2}{4}\cos z \qquad [\text{x.v rule}]$$

$$\frac{D}{D^2+1}(z\cos z) = \frac{z}{2}\sin z - \frac{z^2}{4}\cos z$$

$$PI = -\frac{z^2\cos z}{2} + \frac{z}{2}\sin z - \frac{z^2}{4}\cos z$$

$$PI = \frac{z}{2}\sin z - \frac{3z^2}{4}\cos z.$$

G.S is $y = CF + PI$

$$y = c_1\cos z + c_2\sin z + \frac{z}{2}\sin z - \frac{3z^2}{4}\cos z$$

14, 15, 16

Sir did not told the concept.

(17) A Condenser of capacity C discharged through an inductance L and resistor R in series and the charge q at any time t satisfies the equation. $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = 0$. If $L = 0.25\ H$ $R = 250\ \Omega$ and $C = 2 \times 10^{-6}\ F$. Then the find the charge q at any time $t > 0$, given that at initially charge $q = 0.002$ Couloumbs and current $\frac{dq}{dt}$ is zero in the circuit.

Sol: Given $L = 0.25$ $\quad q_0 = 0.002$ at $t = 0$,

$R = 250$

$C = 2 \times 10^{-6}$ $\quad i = \frac{dq}{dt} = 0$ at $t = 0$

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = 0$$

$$(0.25)\frac{d^2q}{dt^2} + 250\frac{dq}{dt} + \frac{q}{2 \times 10^{-6}} = 0$$

multiply entire eqn with 4.

$$\frac{d^2q}{dt^2} + 1000\frac{dq}{dt} + 2000000\,q = 0$$

operator form $(D^2 + 1000D + 2000000)\,q = 0$

auxillary form $(m^2 + 1000m + 2000000) = 0$

$$m = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$m = \frac{-1000 \pm \sqrt{1000000 - 4(2000000)}}{2}$$

$$m = \frac{-1000 \pm \sqrt{-7,000,000}}{2}$$

$$m = \frac{-500 \pm \sqrt{700 \times 10,000}}{2}$$

$$m = -500 \pm \frac{\sqrt{700 \times 10,000}\cdot i}{2} = -500 \pm 50\sqrt{700}\, i$$

$$= -500 \pm 500\sqrt{7}\, i$$

$$q(t) = e^{-500t}\left[C_1 \cos 500\sqrt{7}\, t + C_2 \sin 500\sqrt{7}\, t\right]$$

at initial conditions, $t = 0$, $q = 0.002$

$$0.002 = e^{0}\left[C_1 \cos(0) + C_2 \sin(0)\right] \Longrightarrow$$

$$C_1 = 0.002$$

at $t = 0$, $\frac{dq}{dt} = 0$

$$\frac{dq}{dt} = -500\, e^{-500t}\left[C_1 \cos(500\sqrt{7}\, t) + C_2 \sin(500\sqrt{7}\, t)\right] + e^{-500t}\left[-500\sqrt{7}\, C_1 \sin(500\sqrt{7}\, t) + 500\sqrt{7}\, C_2 \cos(500\sqrt{7}\, t)\right]$$

at $t = 0$;

$$0 = -500(C_1) + 500\sqrt{7}\,(C_2)$$

$$500\sqrt{7}\,C_2 = 500(0.002)$$

$$B \Rightarrow \frac{0.002}{\sqrt{7}}$$

$$q(t) = e^{-500t}\left[0.002\cos(500\sqrt{7}\,t) + \frac{0.002}{\sqrt{7}}\sin(500\sqrt{7}\,t)\right]$$

$$q(t) = 0.002\, e^{-500t}\left[\cos(500\sqrt{7}\,t) + \frac{1}{\sqrt{7}}\sin(500\sqrt{7}\,t)\right]$$

Coloumb